

Weekly Assignment: Variables, Sets, Relations, and Functions (Epp 1.1, 1.2, & 1.3)

Complete each of the following problems. Show all work and justify your answers.

Problem 1 (Epp 1.1 #2). In each part, fill in the blanks using a variable or variables to rewrite the given statement.

Is there an integer that has a remainder of 2 when it is divided by 5 and a remainder of 3 when it is divided by 6?

- a. Is there an integer n such that n has _____?
- b. Does there exist _____ such that if n is divided by 5 the remainder is 2 and if _____?

Note: There are integers with this property. Can you think of one?

Problem 2 (Epp 1.1 #4). In each part, fill in the blanks using a variable or variables to rewrite the given statement.

Given any real number, there is a real number that is greater.

- a. Given any real number r , there is _____ s such that s is _____.
- b. For any _____, _____ such that $s > r$.

Problem 3 (Epp 1.1 #7 b,d). Rewrite the following statements less formally, without using variables. Determine, as best as you can, whether each statement is true or false.

- b. There is a real number x such that $x^2 < x$.
- d. For all real numbers a and b , $|a + b| \leq |a| + |b|$.

Problem 4 (Epp 1.2 #2 b,d). Write in words how to read each of the following out loud.

- b. $\{x \in \mathbf{R} \mid x \leq 0 \text{ or } x \geq 1\}$
- d. $\{n \in \mathbf{Z}^+ \mid n \text{ is a factor of } 6\}$

Problem 5 (Epp 1.2 #4). Answer each of the following questions. Give reasons for your answers.

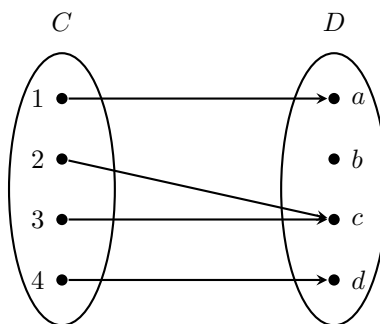
- a. Is $2 \in \{2\}$?
- b. How many elements are in the set $\{2, 2, 2, 2\}$?
- c. How many elements are in the set $\{0, \{0\}\}$?
- d. Is $\{0\} \in \{\{0\}, \{1\}\}$?
- e. Is $0 \in \{\{0\}, \{1\}\}$?

Problem 6 (Epp 1.3 #2). Let $C = D = \{-3, -2, -1, 1, 2, 3\}$ and define a relation S from C to D as follows: For every $(x, y) \in C \times D$,

$$(x, y) \in S \quad \text{means that} \quad \frac{1}{x} - \frac{1}{y} \text{ is an integer.}$$

- Is $2 S 2$? Is $(-1) S (-1)$? Is $(3, 3) \in S$? Is $(3, -3) \in S$?
- Write S as a set of ordered pairs.
- Write the domain and co-domain of S .
- Draw an arrow diagram for S .

Problem 7 (Epp 1.3 #14). Let $C = \{1, 2, 3, 4\}$ and $D = \{a, b, c, d\}$. Define a function $G: C \rightarrow D$ by the following arrow diagram:



- Write the domain and co-domain of G .
- Find $G(1)$, $G(2)$, $G(3)$, and $G(4)$.

Problem 8 (Proof). Prove that for any equation of the form $y^A + x^B = C$, where A , B , and C are positive integers, whether or not the relation corresponding to this equation is a function (with x as input and y as output, over the reals) depends only upon whether A is even or odd.

Hint: Consider two cases based on the parity of A . For each case, think about how many real values of y satisfy $y^A = C - x^B$ for a given x . When does a real number have a unique real A th root, and when might it have zero or two?